AI FACTORY / INCIDENT REPORT

Synthetic bearing line | Educational decision support

Incident f897fc48-b2e7-41f2-a8d2-cef779120e3a

Created 2026-09-06T10:05:54.740228+00:00

Supervisor status: APPROVED

Sensor observation time: 2026-01-01 05:00:00+00:00

Machine M-01 | Failure probability: 0.0% in next 6 hours | Model: random_forest v1

Image classification: normal | Confidence: 99.5% | Domain: synthetic bearing surfaces

Prediction evidence

One-feature replacement with training median (local sensitivity; not causal)

temperature: value 65.033, reference 70.505, probability change -0.000

temperature_mean_6: value 60.123, reference 70.371, probability change -0.000

vibration: value 1.745, reference 3.360, probability change +0.000

pressure: value 6.602, reference 5.939, probability change +0.000

rpm: value 1689.911, reference 1670.118, probability change +0.000

Maintenance note: Bearing inspection with vibration and quality review.

Retrieved evidence

bearing_sop.txt

SYNTHETIC TRAINING MANUAL — not an industrial procedure. Section 1: Vibration and bearing wear Persistent vibration accompanied by rising temperature may indicate bearing wear. Ask the supervisor to inspect the bearing and lubrication history. Compare multiple sensor samples before recommending maintenance. Only qualified personnel may perform maintenance.

bearing_sop.txt

Section 2: Surface quality Scratches and pits on bearing surfaces require supervisor review and segregation of the affected lot for inspection. Investigate whether quality loss coincides with vibration changes. The image classifier is trained on synthetic surfaces only.

bearing_sop.txt

Section 3: Safe maintenance Before maintenance, a qualified supervisor must authorize a stop and apply the site's verified isolation and lockout procedure. This synthetic manual is not a substitute for the actual machine manual. Never execute a stop solely on an AI prediction.

Digital twin comparison

Action	Good units	Downtime h	Expected cost
continue	904.0	0.0	224.0
reduce_load	632.8	0.0	1308.8
maintenance	678.0	2.0	1578.0

Illustrative assumptions: 8-hour horizon; 120 units/hour; failure repair cost 2,400; failure downtime 4 hours; lost-unit cost 4. Planned maintenance takes 2 hours and costs 450. Reduced load uses 70% throughput and 0.55 risk multiplier; maintenance uses 0.15 risk multiplier.

The 6-hour model probability converts to the scenario horizon using a hypothetical constant hazard. Reject rate: 5.8% (production record). Image confidence is not used as the batch reject rate. Monetary amounts are illustrative currency units.

Recommendation: continue

Lowest expected cost among three illustrative scenarios; human approval required.

Human decision

Decision: approve

Final action: continue

Reason: SYNTHETIC REHEARSAL, not a real supervisor decision

Audit ID: fb8978a9-c903-4daf-b3d9-2ef836b2d313

Limitations

Models, images, sensor histories and manuals in this demo are synthetic. Probabilities are uncalibrated; the digital twin uses hypothetical costs and risk multipliers. Grad-CAM and feature sensitivity are explanations of model behavior, not proof of physical cause. No machinery commands are executed.